BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 282 · Network types, architecture and hardware

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does an OS device driver do?

Q2. What does a router connect?

Q3. Define a client-server relationship.

Q4. Binary search on sorted indexes 0–14 for 13: show midpoint index sequence.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LAN — Local Area Network.

WAN — Wide Area Network.

switch — Forwards frames inside local network.

router — Forwards packets between IP networks.

access point — Provides wireless connection to LAN.

New learning

Understand — A LAN covers a local area such as a school building; a WAN connects networks across wider geography.

Trace — Client-server uses dedicated service systems; peer-to-peer allows peers to exchange roles and resources, with different management implications.

Explain — NICs connect devices, switches forward local Ethernet frames, wireless APs bridge Wi-Fi, and routers forward IP packets between networks.

Worked example

Trace school laptop reaching an external site through a wired school network.

1. Predict — Switch moves frames within LAN

2. Check the answer — router handles packets destined for another network.

We Do · Practise with guidance

Two classrooms use Wi-Fi but need access to a central school file server. Identify useful hardware/architecture.

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Clients → access points → switch → file server.
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Predict

APs provide wireless access

Check the answer

switch connects local devices

client-server centralises authentication/files.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define LAN, WAN, switch and router. [4]

Question 2

Describe a laptop's network path via wireless access point, switch and router to the Internet. [6]

Question 3

Compare client-server and peer-to-peer for school file sharing, giving strengths and limits. [6]

Exit Ticket · Answer before checking

A network switch and router are always interchangeable because they both move packets.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Offers an interface to hardware.

Q2. Different IP networks.

Q3. Client requests service from a dedicated server.

Q4. mid7, mid11, mid13; target found at index13.

You Do pages · Independent questions

Q1. LAN local network [1], WAN wide-area connection [1], switch forwards LAN frames [1], router joins IP networks [1].

Q2. NIC/Wi-Fi interface [1], AP receives frame [1], switch forwards on LAN [1], router selects external network [1], ISP/WAN carries traffic [1], return follows appropriate route [1].

Q3. Client-server central storage/permission policies [2] but server dependency/cost [1]; P2P direct resource sharing [2] but decentralised management [1].

Exit Ticket · Model answer

False. Their usual layer and forwarding responsibilities differ.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding